MARWAN TAREK

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SUMMARY

Software Engineer with 3+ years of experience in **Python (FastAPI)**, **React/TypeScript**, **C#/.NET** and **C++**, building cloud-native **microservices** and **Generative AI** applications, including Retrieval-Augmented Generation (**RAG**) and multi-agent **LLM** pipelines, on **PostgreSQL**, **MongoDB** and **Redis**. Hands-on with **Kubernetes**, **Docker**, **Azure**, **Kafka**, **Keycloak**, REST APIs, CI/CD, automated testing and observability.

EXPERIENCE

Alstom — Software Engineer Jul 2025 – Present *Cairo, Egypt*

- Designed and deployed **Python/FastAPI microservices** and **React/TypeScript** UIs for railway signalling (CBTC, Automatic Train Supervision) on **Kubernetes**, with versioned **OpenAPI** contracts.
- Orchestrated **multi-agent LLM** pipelines and **RAG** services (hybrid BM25 and vector search, LLM-as-judge) on self-hosted models via **vLLM** to generate specifications and trace requirement coverage.
- Created an **RCA Copilot** for root-cause analysis using **Azure OpenAI**, **RAG**, **ChromaDB/Qdrant** and a React chat UI, improving incident triage and reducing mean time to detect.
- Engineered a real-time dashboard with FastAPI workers, **Redis Streams** and SSE feeding React, backed by **TimescaleDB** and **Keycloak** (OIDC), and scaled horizontally with Redis pub/sub and failover.
- Developed idempotent, back-pressure-aware event-driven services on **Kafka** and **ActiveMQ Artemis**, including a **C#/.NET** service; implemented **TLS**, **JWT**, **RBAC** and **OpenTelemetry** distributed tracing.
- Automated deployment with **Docker**, **Rancher**, **Harbor** and **Azure** via **GitLab CI/CD** on **RHEL**; added **Robot Framework** tests and quality gates, raising regression coverage by about **30%**.

Spark Systems — Software Engineer (.NET) Jan 2025 – Jun 2025 *Giza, Egypt*

- Developed **REST APIs** and business logic using **C#/.NET Core**, **Entity Framework** and **Clean/Onion Architecture**; documented API contracts with **OpenAPI** and validated workflows using **Postman**.
- Implemented authentication and role/claims-based authorization with **.NET Identity**, plus **repository** and **unit-of-work** patterns.
- Designed **PostgreSQL** persistence, tuned **SQL queries** and added **Redis caching** for performance; shipped with **Docker** and **Jenkins CI/CD** in Agile/Scrum with code reviews.

Valeo — Embedded Software Engineer Jul 2022 – Oct 2023 *Smart Village, Egypt*

- Developed & validated **multithreaded C/C++** bootloader, memory and **CAN/SPI/UART** drivers on an **RTOS**; applied **traceability**, **coverage analysis** and **static analysis**.

CORE SKILLS

Languages: Python, C#, C/C++, TypeScript, JavaScript, SQL, Bash/Shell
Backend & APIs: FastAPI, Flask, Node.js/Express, EF Core, GraphQL, gRPC, OpenAPI/Swagger
Frontend & Testing: React, SignalR, WebSockets; PyTest, Playwright, xUnit, NUnit, Robot Framework
AI & LLM: Generative AI, AI Agents, RAG, Prompt Engineering, Embeddings, vLLM, Azure OpenAI
Data & Messaging: PostgreSQL, SQL Server, MongoDB, Redis, InfluxDB, Qdrant, Kafka, ActiveMQ
Cloud & DevOps: Azure (App Service, Functions, API Management, Key Vault, Container Registry, VMs), Docker, Kubernetes, Terraform, Bicep, Ansible, GitLab CI, GitHub Actions, Jenkins, Git, Linux (Red Hat)
Architecture: Microservices, DDD, Clean/Onion, CQRS, SOLID, Design Patterns, Resilience (Polly)
Security & Monitoring: OAuth 2.0/OIDC, Keycloak, JWT/Cookie Auth, RBAC, TLS, Prometheus, Grafana
Fundamentals: OOP, Data Structures & Algorithms, Socket Programming, Unit Testing, Debugging

EDUCATION

B.Sc. in Computer Science (Honours), Helwan University Sep 2019 – Jul 2023
Languages: English (C2), French (A2), Arabic (Native)